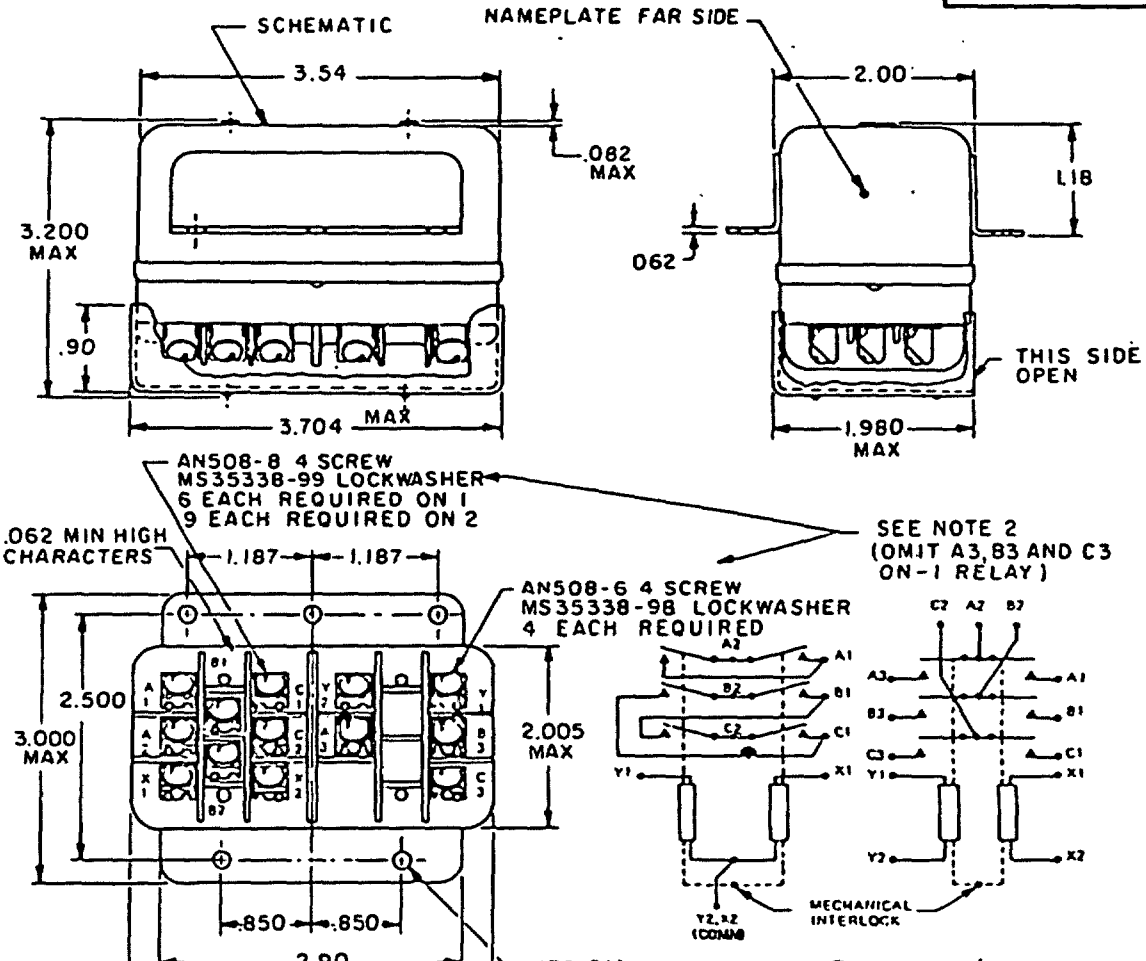
THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

THIS SPECIFICATION IS APPROVED FOR USE BY all Departments and Agencies of the Department of Defense.

Form Approved
OMB No. 0704-0188



CIRCUIT PART NOS.-1B (3 POLE MOTOR REVERSING) Ⓐ

CIRCUIT PART NOS.-2B (LOAD TRANSFER) Ⓐ

Inches	mm	Inches	mm
.062	1.57	2.005	50.93
.082	2.08	2.500	63.50
.173	4.39	2.90	73.66
.850	21.59	3.000	76.20
.90	22.86	3.200	81.28
1.18	29.97	3.54	89.92
1.187	30.15	3.671	93.24
1.980	50.29	3.704	94.08
2.00	50.80		

NOTES:
1. Dimensions are in inches unless otherwise specified, tolerances are decimals-XX ±.030, XXX ±.010.
2. Power terminals capable of accepting MS25036-53 lugs. Coil terminals capable of accepting MS25036-7 lugs.
3. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

Ⓕ Denotes Changes

PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - NAVY - AIR FORCE - DLA -
REVIEW: Air Force-99
USER:
PROJECT NUMBER: 5945-F712

MILITARY SPECIFICATION SHEET
TITLE RELAY, ELECTROMAGNETIC, 20 AMPERES, 3 PST, MECHANICAL INTERLOCK, (CENTER OFF) LOAD TRANSFER, TYPE I, HERMETICALLY SEALED.

SPECIFICATION SHEET NUMBER
MS27706F 16 March 1990
SUPERSEDING
MS27706E 30 April 1987
AMSC - N/A FSC - 5945

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

DD Form 672, MAY 88 Previous editions are obsolete.
* U.S. GOVERNMENT PRINTING OFFICE: 1988-740-979:85979

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:48Z
Check the source to verify that this is the current version before use.

THIS SPECIFICATION IS APPROVED FOR USE BY all Departments and Agencies of the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106

(F) TABLE I. Dash number and characteristics.

MS part no. MS27706	Type	Coil X	Coil Y	Terminal types	Mounting	Max weight in pounds
-1B -2B	I	dc	dc	Screw	Bracket	1.4

(F) TABLE II. Operating characteristics.

MS part no. MS27706	Coil data: Coil	Nominal: Volts 1/	Nominal: Freq Hz	Nominal: Res Ω	Max: Volts	Max: Amp.	Max pick-up voltage 2/: Nominal 2/	High temp test	Cont current test	Drop-out voltage 2/	Hold voltage 2/	Time-milliseconds-max: Operate 3/	Release 4/	Bounce: Main NO	Main NC	Aux NO	Aux NC
-1B, -2B	X-Y	28	dc	-	29	.25	18	19.5	22.5	1.5	7.0	20	10	5	-	-	-

1/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

Form Approved OMB No. 0704-0188

PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - NAVY - AIR FORCE - DLA -
REVIEW: Air Force-99
USER:
PROJECT NUMBER: 5945-F712

MILITARY SPECIFICATION SHEET
TITLE: RELAY, ELECTROMAGNETIC, 20 AMPERES, 3 PST, MECHANICAL INTERLOCK, (CENTER OFF) LOAD TRANSFER, TYPE I, HERMETICALLY SEALED.

SPECIFICATION SHEET NUMBER: 16 March 1990
SUPERSEDING: MS27706E 30 April 1987
AMSC - N/A FSC - 5945

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Page 2 of 3

DD Form 672, MAY 88 Previous editions are obsolete.

* U.S. GOVERNMENT PRINTING OFFICE: 1988-240-979:60928

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:48Z
Check the source to verify that this is the current version before use.

THIS SPECIFICATION IS APPROVED FOR USE BY all Departments and Agencies of the Department of Defense.

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(s) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-R-6106.

Table III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc 1/				115 V ac, Phase				115/200 V ac, 3 Phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	20	-	-	-	20	15	-	-	20	15	-	-	
Inductive	25	15	-	-	-	-	-	-	-	-	-	-	-	
Inductive	100	-	-	-	-	20	20	-	-	20	20	-	-	
Motor	100	20	-	-	-	20	12	-	-	20	12	-	-	
Lamp	100	5	-	-	-	5	5	-	-	5	5	-	-	
Transfer load	10	-	-	-	-	20	-	-	-	20	-	-		2/
Mechanical life Reduced current	400	5	-	-	-	5	5	-	-	5	5		-	2/
Min current		Applicable per specification												

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplied at rating indicated.

Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 Ft
Shock G-level	25 G
Duration	11 ms
Max duration contact opening	2 ms
Vibration:	
Sinusodial:	
G-level	10 G
Frequency range	10-1500 Hz
Non operate:	
G-level	15 G
Frequency range	20-2000 Hz
Acceleration	15 G

Electrical characteristics

Insulation resistance initial		100 megohms
After life or environmental test		50 megohms
Dielectric strength (sea level):		
	Initial	After life tests
Coil to case	1250 V rms	1050 V rms
Aux contacts	---	---
All other points	1500 V rms	1125 V rms
Dielectric strength (altitude)		
	80,000 Ft.	
Coil to case	500 V rms	
Aux contacts	---	
All other points	500 V rms	
Max contact drop initial		.150 volts
After life test		.175 volts
Overloading current	80 Amperes dc, 120 amperes ac	
Rupture current	100 Amperes dc, 150 amperes ac	
Duty rating	continuous	
RFI spec	MIL-STD-461	
(Applicable to coil circuit of ac operated relays)		

PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - NAVY - AIR FORCE - DLA -
REVIEW: Air Force-99
USER:
PROJECT NUMBER: 5945-F712
DISTRIBUTION STATEMENT A: Approved for public release; distribution unlimited.

MILITARY SPECIFICATION SHEET
TITLE RELAY, ELECTROMAGNETIC, 20 AMPERES, 3 PST, MECHANICAL INTERLOCK, (CENTER OFF) LOAD TRANSFER, TYPE 1, HERMETICALLY SEALED.

SPECIFICATION SHEET NUMBER
SUPERSEDING MS 27706E
AMSC N/A
FSC 5945

16 March 1990
30 April 1987

Page 3 of 3

DD Form 672, MAY 88
Previous editions are obsolete.
* U.S. GOVERNMENT PRINTING OFFICE: 1988-242-070/80328

Form Approved OMB No. 0704-0188

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:48Z
Check the source to verify that this is the current version before use.